

# WEEK 6 – POWER BI HEALTHCARE DATA REPORT

## Healthcare Analytics Dashboard – DAX Measures & Interactive Visualizations

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### Introduction

This week focuses on creating DAX measures and developing an interactive healthcare analytics dashboard using the cleaned healthcare dataset. The dashboard provides insights into patient counts, billing amounts, length of hospital stay, medical conditions, admission types, insurance providers, and admission trends.

### Objectives

- To create and use **DAX measures** for healthcare data analysis.
- To calculate key metrics such as **Total Patients, Total Billing Amount, Average Billing Amount, and Average Length of Stay**.
- To analyze patient demographics based on **Age Group and Gender**.
- To analyze **Medical Conditions and Admission Types** using interactive visualizations.
- To examine **admission trends and billing patterns** over time.
- To compare billing amounts across different **Insurance Providers**.
- To create an **interactive healthcare dashboard** using KPI cards, charts, and slicers.
- To enable dynamic data analysis using **Insurance Provider, Admission Type, and Date Range** filters.

## **STEP 1 – Create Total Patients Measure**

A DAX measure was created to calculate the total number of patient/admission records in the `Billing` table.

### **DAX Formula**

Total Patients = `COUNTROWS(Billing)`

### **Explanation**

The `COUNTROWS` function counts the number of records available in the `Billing` table.

## **STEP 2 – Create Total Billing Amount Measure**

A measure was created to calculate the total billing amount for all healthcare records.

### **DAX Formula**

Total Billing Amount = `SUM(Billing[Billing Amount])`

### **Explanation**

The `SUM` function calculates the total billing amount from the `Billing` table.

## **STEP 3 – Create Average Billing Amount**

A DAX measure was created to calculate the average billing amount per record.

### **DAX Formula**

Average Billing Amount = `AVERAGE(Billing[Billing Amount])`

### **Explanation**

The AVERAGE function calculates the mean billing amount across the healthcare records.

#### **STEP 4 – Calculate Average Length of Stay**

The average number of days between admission and discharge was calculated using DATEDIFF and AVERAGEX.

##### **DAX Formula**

```
Average Length of Stay =  
AVERAGEX(  
    Billing,  
    DATEDIFF(  
        Billing[Date of Admission],  
        Billing[Discharge Date],  
        DAY  
    )  
)
```

##### **Explanation**

The DATEDIFF function calculates the number of days for each patient's stay, while AVERAGEX calculates the average length of stay.

#### **STEP 5 – Create Readmission Rate**

A measure was created to identify patients having more than one admission and calculate the readmission rate.

##### **DAX Formula**

```
Readmission Rate =  
VAR TotalPatients =  
    DISTINCTCOUNT(Billing[Patient ID])  
VAR RepeatPatients =
```

```

COUNTROWS(
  FILTER(
    SUMMARIZE(
      Billing,
      Billing[Patient ID],
      "AdmissionCount", COUNTROWS(Billing)
    ),
    [AdmissionCount] > 1
  )
)
RETURN
  DIVIDE(RepeatPatients, TotalPatients, 0)

```

### Explanation

The measure identifies patients with more than one admission and calculates their proportion among the total number of patients.

## STEP 6 – Create KPI Summary Cards

Four KPI cards were added to the dashboard to provide a quick summary of the healthcare data.

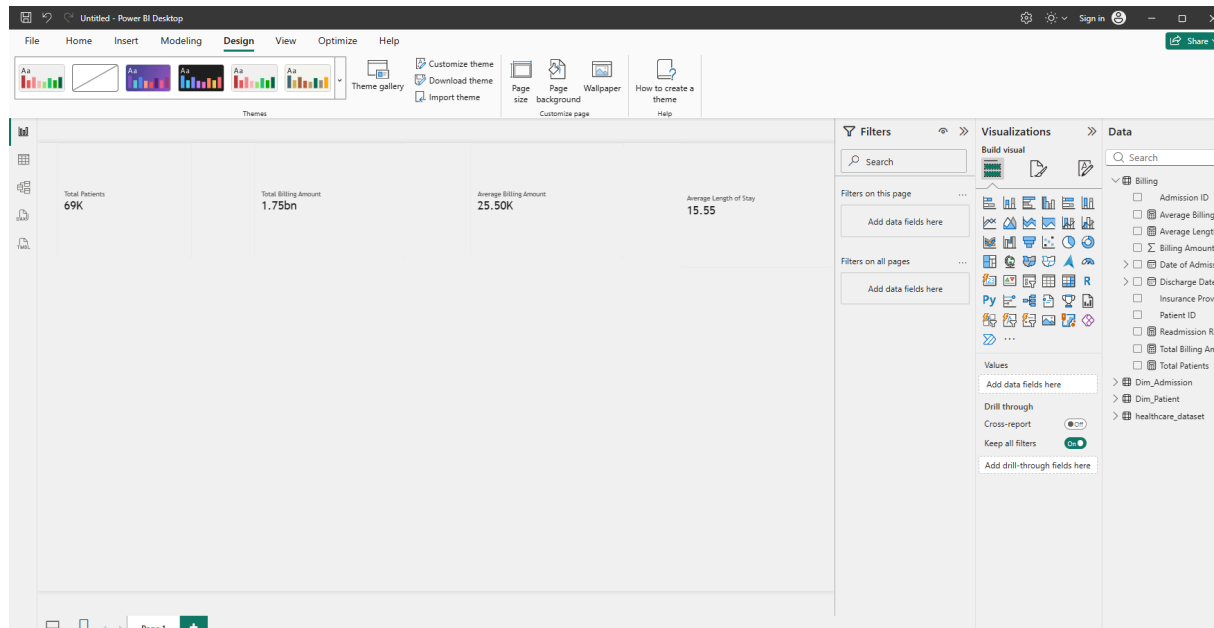
The KPI cards display:

- **Total Patients**
- **Total Billing Amount**
- **Average Billing Amount**
- **Average Length of Stay**

### Explanation

These cards provide an immediate overview of the major healthcare performance indicators.

**Screenshot:** Four KPI summary cards.



## STEP 7 – Create Demographic Matrix

A Matrix visual was created to analyze patient distribution by age group, gender, and medical condition.

### Fields Used

#### Rows:

- Age Group
- Gender

#### Columns:

- Medical Condition

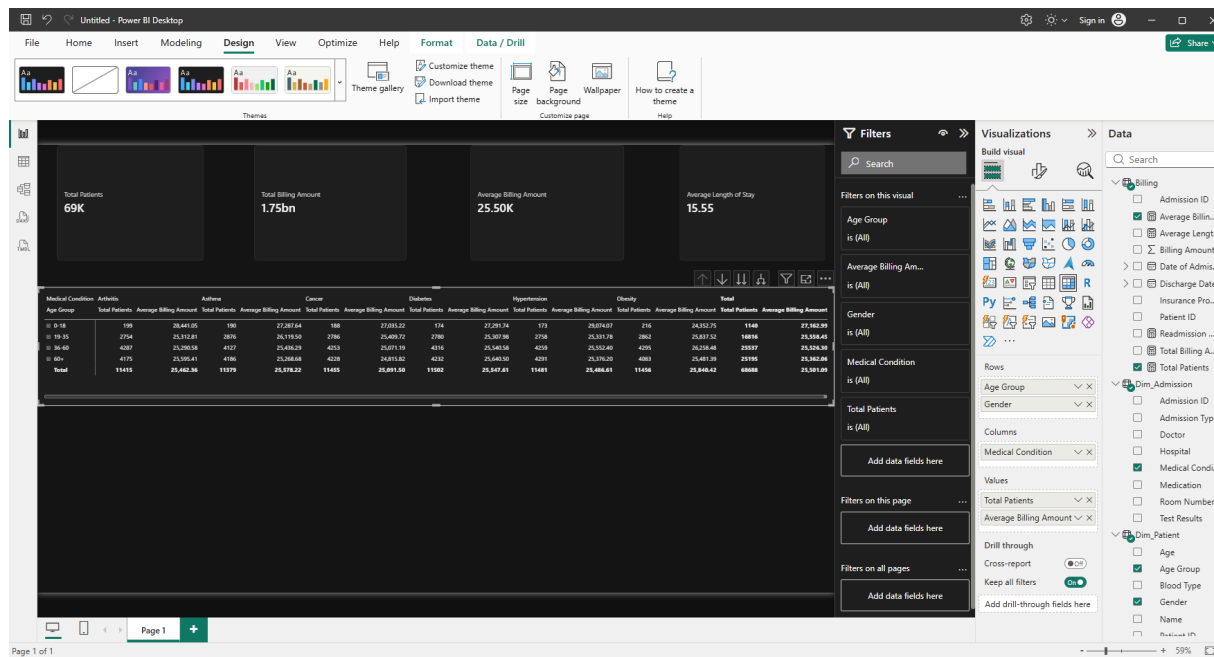
#### Values:

- Total Patients
- Average Billing Amount

### Explanation

The matrix provides a detailed comparison of patient demographics and billing amounts across different medical conditions.

**Screenshot:** Demographics Matrix.



## STEP 8 – Create Age Group Donut Chart

A Donut Chart was created to show the distribution of patients across different age groups.

### Fields Used

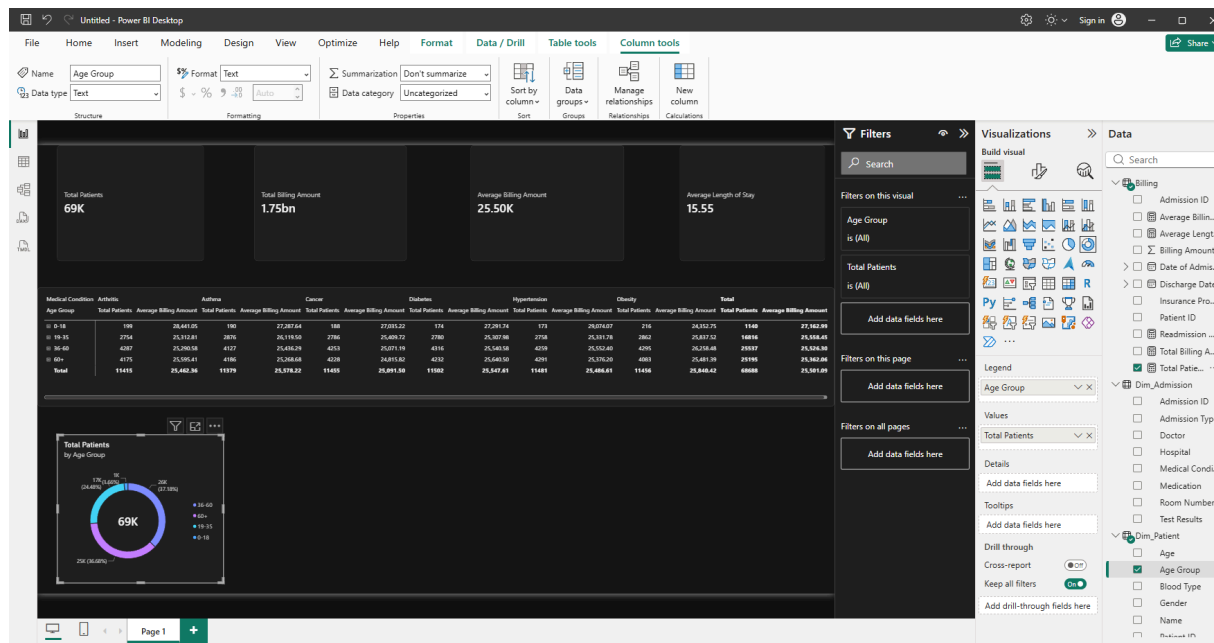
**Legend:** Age Group

**Values:** Total Patients

### Explanation

The donut chart visually represents the proportion of patients in each age category.

**Screenshot:** Age Group Donut Chart.



## STEP 9 – Create Medical Condition & Admission Chart

A Stacked Bar Chart was created to analyze patient volumes across different medical conditions and admission types.

### Fields Used

**Y-Axis:** Medical Condition

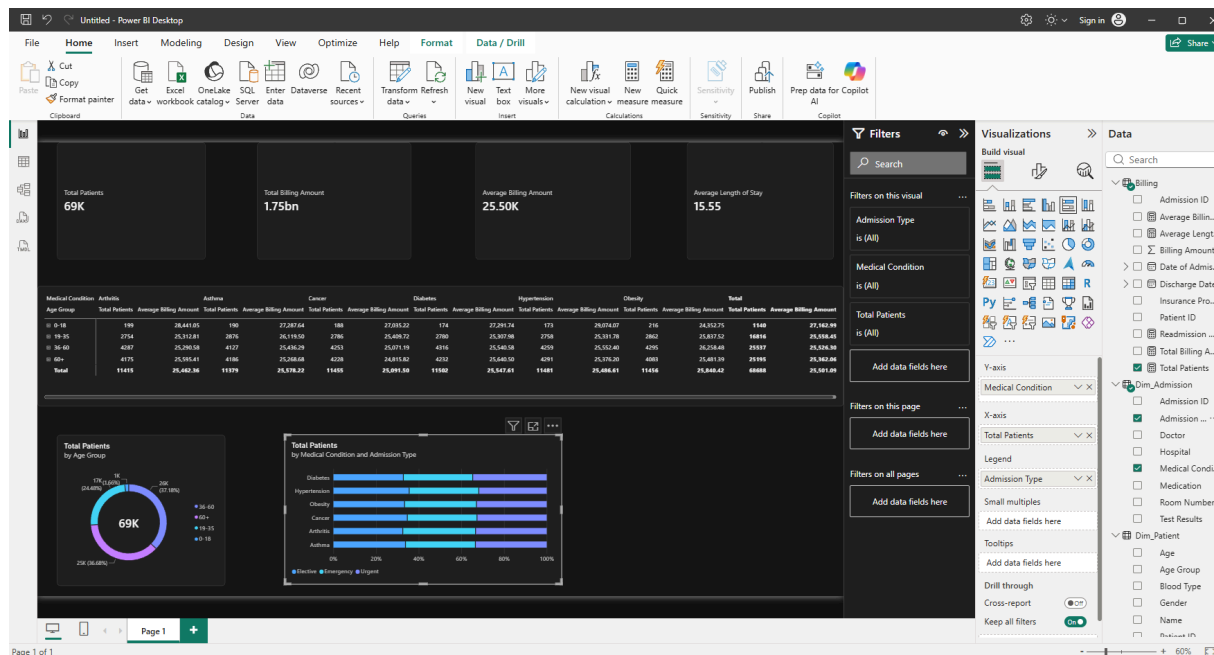
**X-Axis:** Total Patients

**Legend:** Admission Type

### Explanation

The chart compares patient counts for different medical conditions while showing the distribution of Emergency, Elective, and Urgent admissions.

**Screenshot:** Stacked Bar Chart.



## STEP 10 – Create Admission Volume Trend

A Line Chart was created to analyze patient admissions over time.

### Fields Used

**X-Axis:** Date of Admission – Year and Month

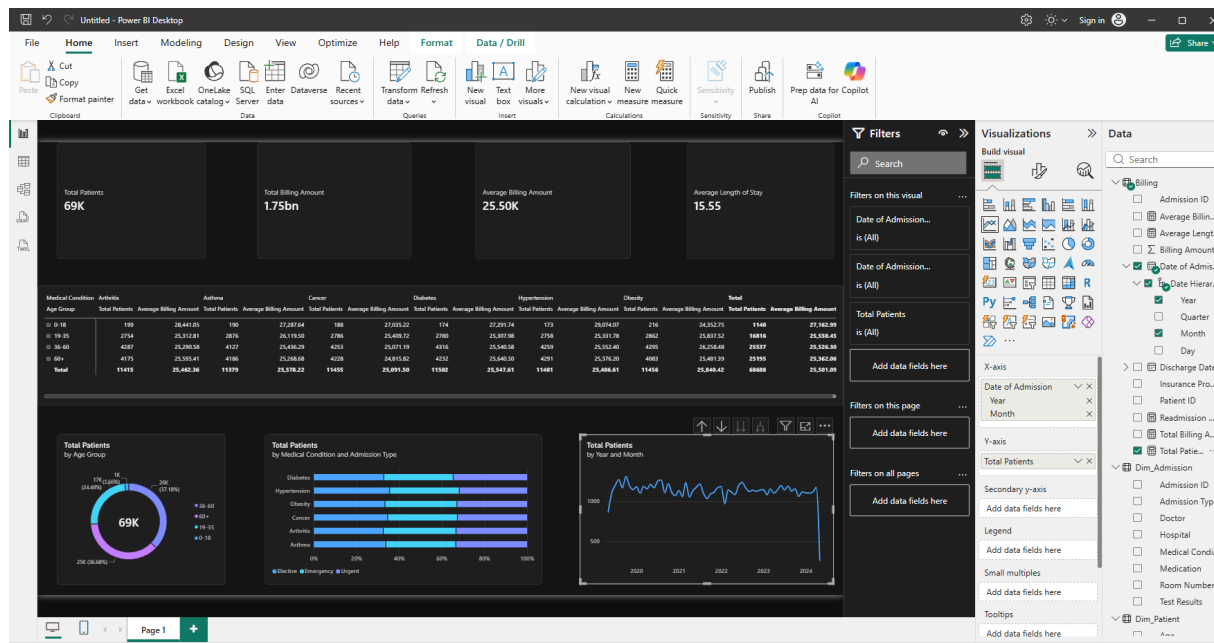
**Y-Axis:**

- Total Patients
- Total Billing Amount

### Explanation

The line chart helps identify changes in patient admissions and billing amounts over the admission period.

**Screenshot:** Admission Volume Line Chart.



## STEP 11 – Create Insurance Provider Billing Chart

A Clustered Column Chart was created to compare billing amounts across insurance providers.

### Fields Used

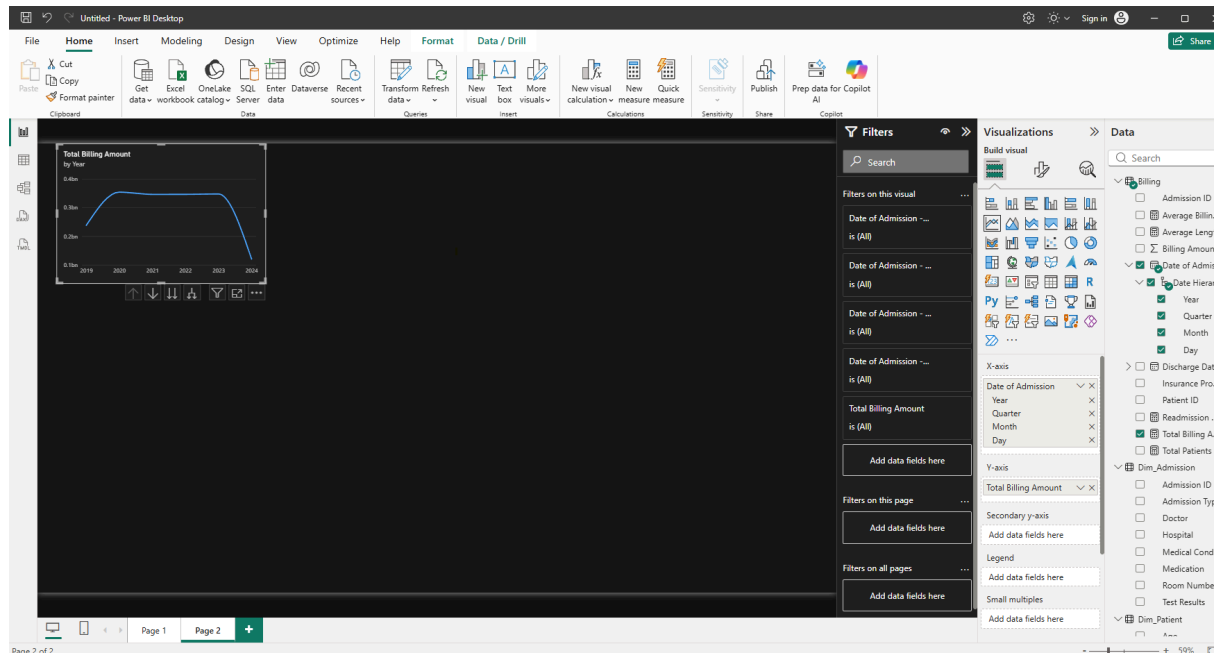
**X-Axis:** Insurance Provider

**Y-Axis:** Total Billing Amount

### Explanation

The chart compares the total billing amount associated with different insurance providers.

**Screenshot:** Insurance Provider Column Chart.



## STEP 12 – Create Insurance Provider Slicer

A slicer was added using the **Insurance Provider** field.

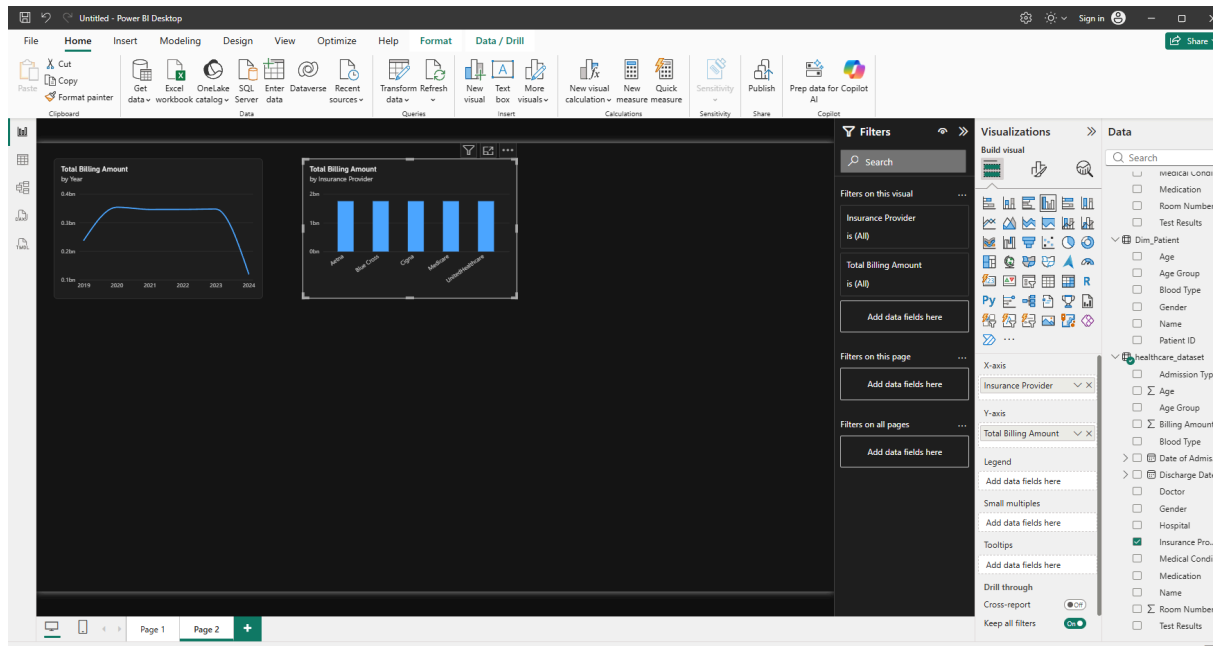
### Field

### Insurance Provider

### Explanation

The slicer allows users to filter the entire dashboard according to a selected insurance provider.

**Screenshot:** Insurance Provider Slicer.



## STEP 13 – Create Admission Type Slicer

A second slicer was added using **Admission Type**.

### Field

### Admission Type

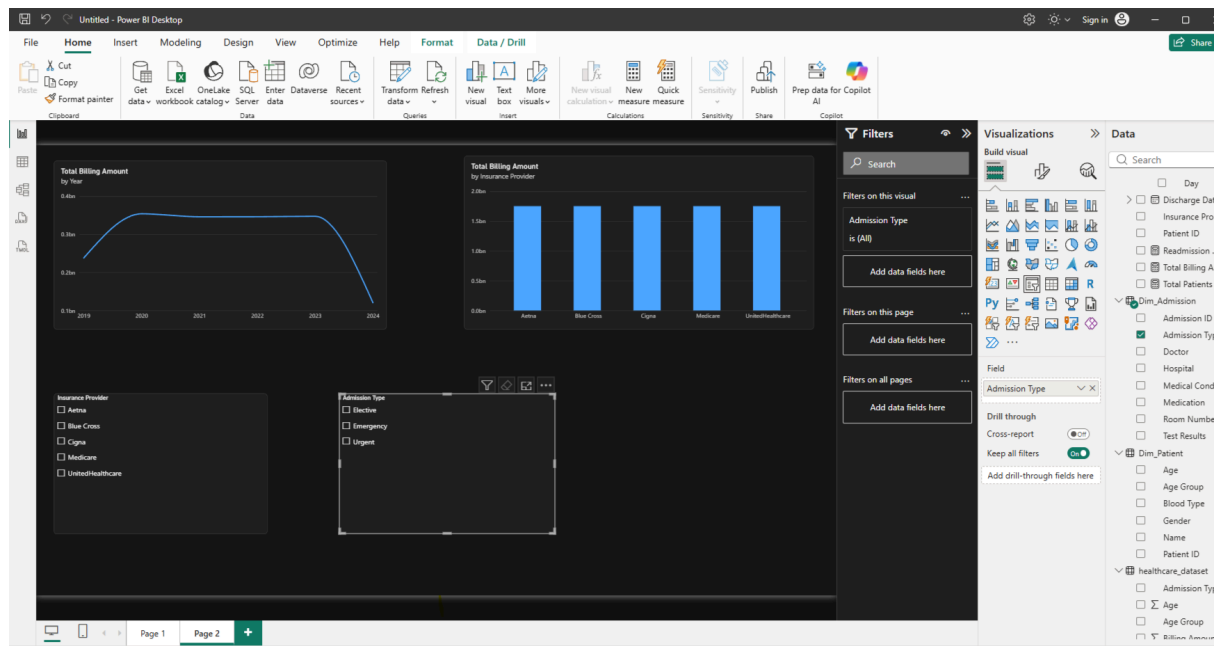
### Available Categories

- Emergency
- Elective
- Urgent

### Explanation

The slicer allows users to dynamically filter the dashboard based on admission type.

**Screenshot:** Admission Type Slicer.



## STEP 14 – Create Date Range Slicer

A date slicer was added using **Date of Admission**.

**Field**

**Date of Admission**

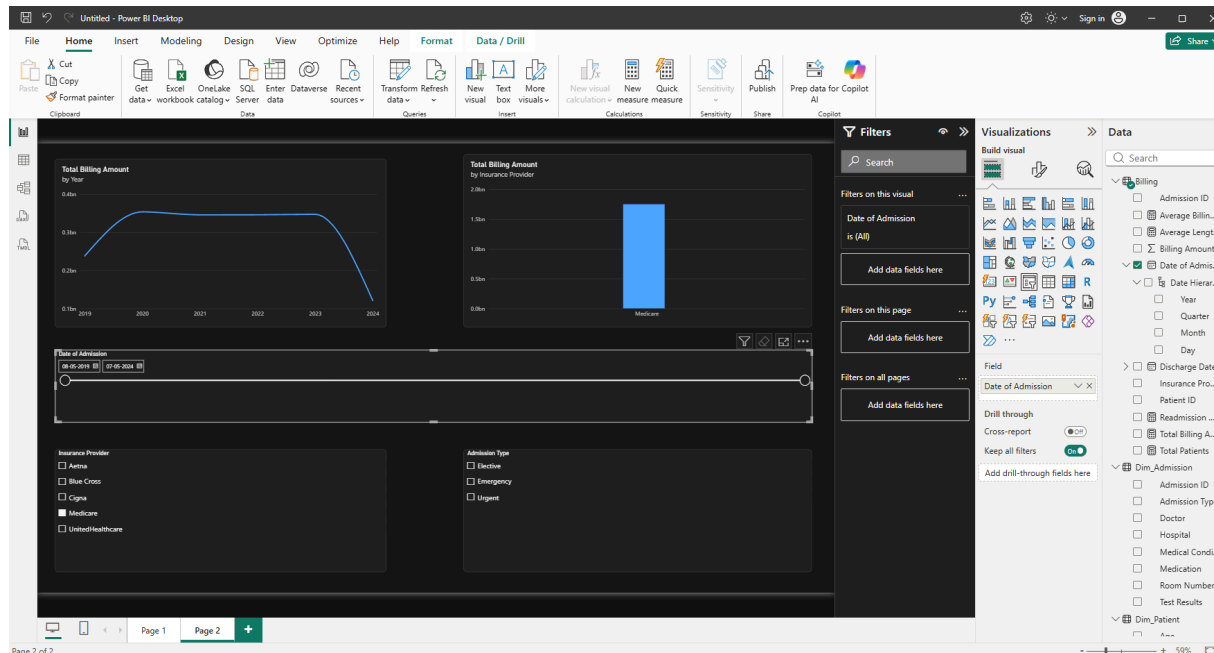
**Slicer Type**

**Between**

**Explanation**

The date range slicer allows users to select a specific admission period and dynamically update the dashboard visuals.

**Screenshot:** Date Range Slicer.



## STEP 15 – Final Healthcare Analytics Dashboard

All KPI cards, charts, matrix, and slicers were arranged into a single interactive dashboard.

### Dashboard Components

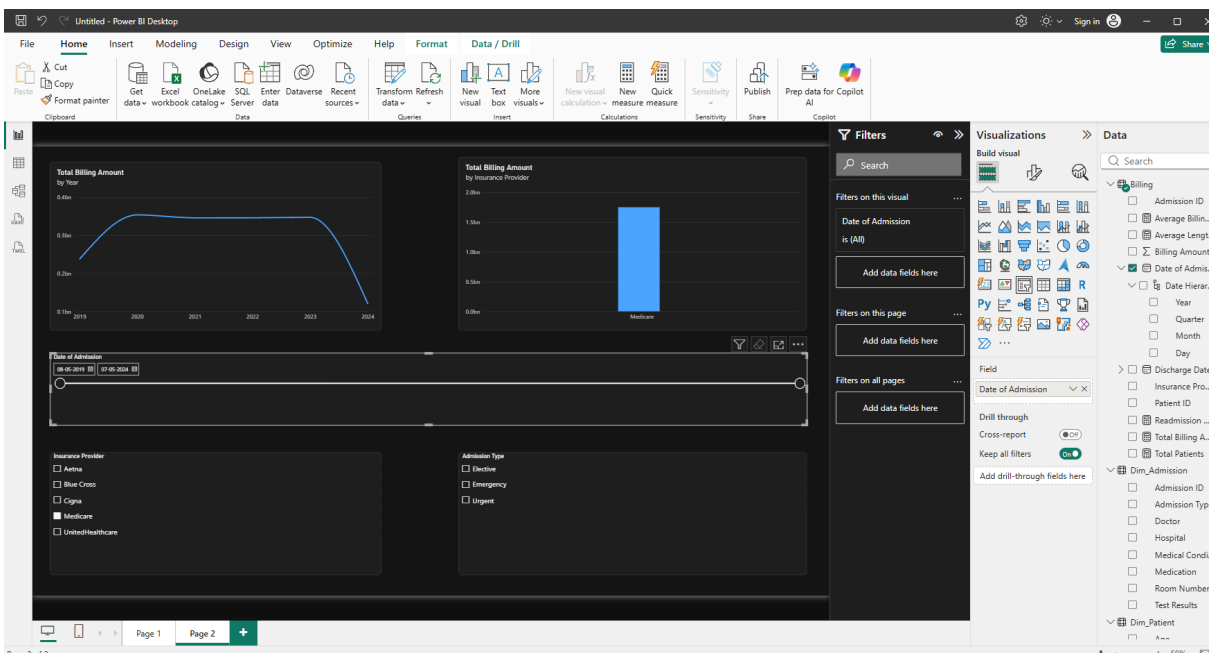
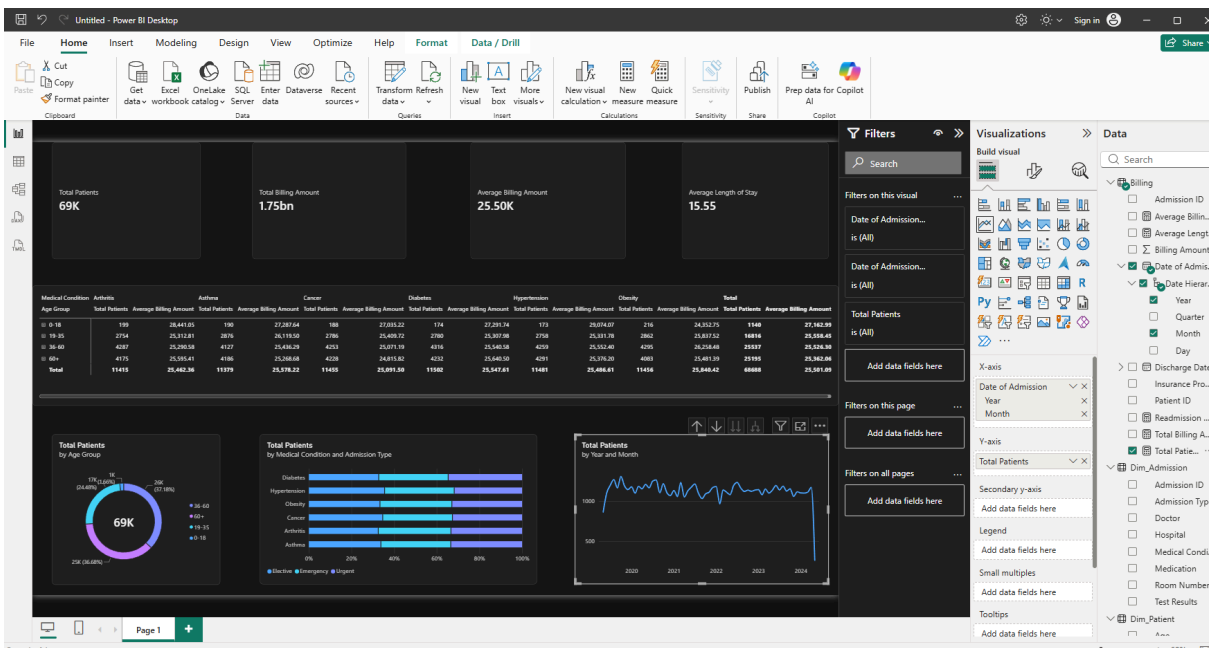
- Total Patients KPI
- Total Billing Amount KPI
- Average Billing Amount KPI
- Average Length of Stay KPI
- Demographic Matrix
- Age Group Donut Chart
- Medical Condition Stacked Bar Chart
- Admission Trend Line Chart
- Insurance Provider Column Chart
- Insurance Provider Slicer
- Admission Type Slicer
- Date Range Slicer

### Explanation

The final dashboard provides an interactive view of healthcare data. Users can use the slicers to filter the visuals and analyze patient

demographics, admissions, billing, medical conditions, and insurance-related information.

Screenshot: Final Healthcare Analytics Dashboard.



## **Conclusion**

The Week 6 Power BI project successfully developed an interactive healthcare analytics dashboard using DAX measures and multiple visualization techniques. The dashboard summarizes patient information, billing performance, admission trends, medical conditions, and insurance provider analysis. The use of slicers enables dynamic filtering and makes the dashboard useful for exploring healthcare data.